

YOURSTORY



H2S



COCO CLI

HACKATHON

GCC Edition



Team Name : Snow Build

Team Leader Name : Vipul Maurya

Team Size : 1

Problem Statement : Supply Chain Ontology and Governed Conversational Analytics

1. Problem Brief

Current challenge: Planning, Procurement and Logistics answer the same question – “What was our on-time delivery for Q3 2026 orders?” – with 83.51%, 96.65% and 64.12%, built from fragmented ERP, logistics, supplier and IoT sources.

Target users: Supply Chain Planner, Procurement Manager, Logistics Manager and Leadership.

Goal: One governed ontology and semantic layer so every persona gets one consistent, traceable answer – and can act on it safely.

Business impact: faster decisions, trusted KPI lineage, early risk detection and governed escalation.

Solution: one trusted answer → continuously defended → simulated & prescribed → acted on

2. End-to-End Workflow (Input -> Processing -> Output)

Input: Natural-language persona questions, contract/SLA lookups and what-if scenarios via CoCo CLI and the Streamlit control tower.

Processing: RAW → CURATED pipelines, governed semantic view, Cortex Agent (Analyst + Search) with guardrails, metric registry + provenance + drift sentinel, risk scoring and recommendation engine.

Output: canonical KPI answers with formula, source and confidence badge; drift status; prioritized recommendations; what-if projections; escalations only after confirmation.

Core stack: SC_ONTOLOGY (RAW, LEGACY, CURATED, SEMANTIC, GOVERNANCE, APP), HACK_WH (XSMALL), Streamlit in Snowflake.

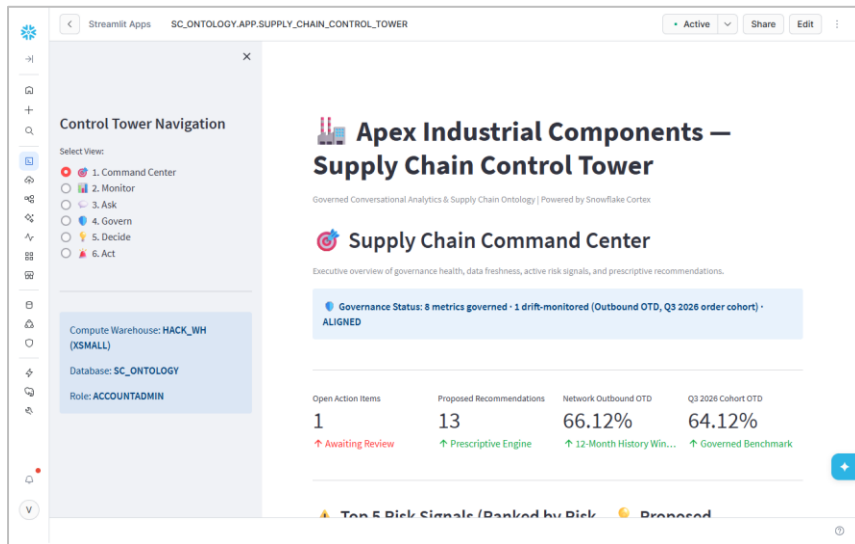
Recorded demo shows explicit CLI Input -> Processing -> Output sequence

3. Fully Working Capabilities

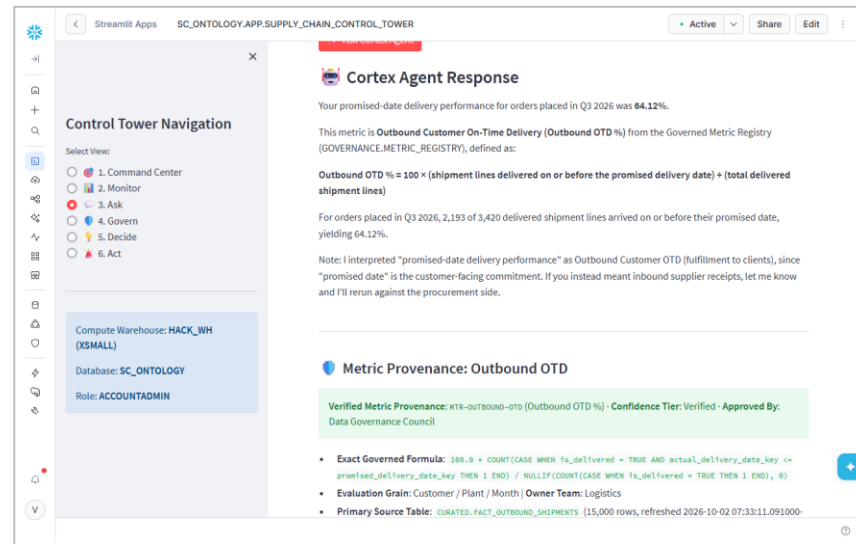
1. **Command Center:** one-screen roll-up of KPIs, governance health and top risks.
2. **Monitor:** governed KPIs with filters + Before-vs-After (3 legacy numbers → one canonical 64.12%).
3. **Ask:** multi-persona chat via Cortex Agent with provenance badge (formula, owner, source, freshness, Verified tier) and guardrails.
4. **Govern:** metric registry glossary, drift sentinel health, ontology & lineage graph.
5. **Decide:** 38 risk signals, 13 prescriptive recommendations, illustrative what-if simulator.
6. **Act:** governed escalation – validated procedure + explicit confirmation; no unintended writes.

Status: Phases 1–9 deployed · Final app validated 8/8 PASS

4. Final App: Command Center & Governed Answers



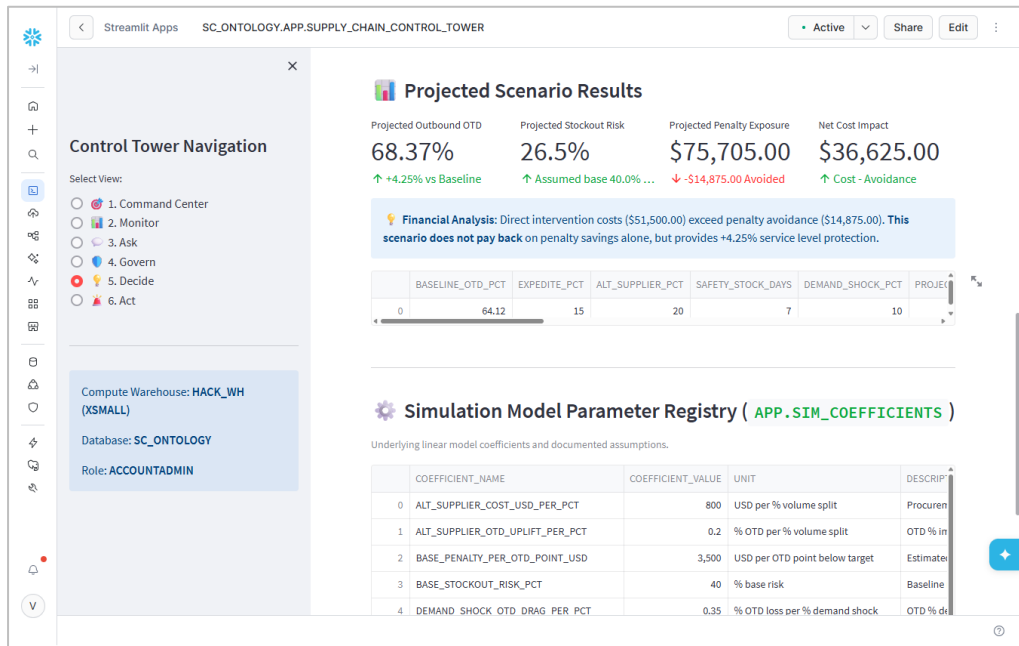
Command Center – KPIs, governance health, top risks



Ask – 64.12% for every persona, with provenance badge

Live in Snowflake: SC_ONTOLOGY.APP.SUPPLY_CHAIN_CONTROL_TOWER

5. Final App: Prescriptive Decisions & What-If



Risk signals: 38 suppliers & carriers scored 0–100; top = SUP-104 (Titan Micro-Foundry).

Recommendations: 13 deterministic actions, e.g. dual-source SUP-104, move lanes away from CARR-01/03/07.

What-if: OTD 64.12% → 68.37%, net cost +\$36,625 → does not pay back.

Honest by design: illustrative heuristics, clearly labelled – not a forecast.

Recommendations become actions only through the governed Act page

6. How CoCo CLI Was Used: Plan → Build → Test

Phase	What CoCo did	Result
Plan	Clarifying questions, region/model checks, blueprint (PLAN.md)	Approved phased plan
1 Data	DB, schemas, synthetic data with injected issues; self-fixed SQL errors	Legacy 83.51 / 96.65 / 64.12
2 Pipelines	Dynamic tables, stream + tasks, data-quality checks	All checks PASSED
3 Semantic	Semantic view + 10 verified queries	Canonical OTD 64.12%
4 AI	Cortex Search, Cortex Agent, validated escalation procedure	3 personas → 64.12%
5–6 Prototype	Streamlit app, deployment, CLI test suite	5 / 5 PASS
7 Governance	Metric registry, provenance, drift sentinel, guardrails	ALIGNED · Verified
8 Decisions	Risk signals, recommendations, what-if function	38 signals · 13 recs
9 Final app	6-page app, deploy, browser smoke test, UI fixes	6 / 6 pages PASS
10 Testing	Read-only validation suite incl. negative test	8 / 8 PASS

Every SQL step was reviewed and approved in CoCo CLI before execution

Prototype: 5/5 PASS · Browser smoke test of the final app: 6/6 pages PASS.

[illegible]

Evidence: final app deploy validation (left) · 8/8 test scorecard (right), both in CoCo CLI

8. Submission Artifacts

GitHub repository: github.com/mauryavipul/apex-supply-chain-ontology-coco

In the repo: SQL 01–23 in run order, app source (prototype + final), PLAN.md (Phases 1–10), prototype & final app validation summaries, CoCo build journey (PDF), screenshots.

Demo video: prototype recording (link in the repo README).

Live app: SC_ONTOLOGY.APP.SUPPLY_CHAIN_CONTROL_TOWER – Streamlit in Snowflake (requires account access).

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THANK YOU

